

Assignment(ERD Design)

Assignment-01:

You have to develop a **Human Resource Management** database for a company. It requires that you have to add some information of departments, employees and projects including company. As mentioned that every department has many employees and each employee work for a department and each department is leading by only one manager who is also an employee. Initially a new department need not have any employee. Here, though an employee belongs a department but they can work for different projects at the same time. Each employee can work for different job position like Developer, Programmer and front-end designer.

Steps of Constructing ERD

1. Identify the Entities Required
2. Identify the Attributes and Primary key for each Entity
3. Identify the Relationship needed
4. Identify the Cardinality Ratio and Participation
5. Draw the Diagram

Step 1: Identify Entities

We have to develop a database for a company including some information of departments, employees and projects of the company. As mentioned that every department has many employees and each employee work for a department and each department is leading by only one manager who is also an employee. Initially a new department need not have any employee. Here, though an employee belongs a department but they can work for different projects at the same time.

Step 2: Identify Attributes and Primary Key Entity

- **Company** (Id, company_name, location, founded_date, managing_director)
- **Departments** (Department_id, name, number_of_employee, manager_id, employee_id, company_id)
- **Employees** (id, first_name, middle_name, last_name, joining_date, email, phone, date_of_birth, gender, designation, salary, department_id, job_title)
- **Address** (Id, employee_id, Village_name, post_office, post_code, Upazila, District)
- **Phone** (Id, number, code, employee_id, sim_operator) o
Job Position (position_id, position_name, employee_id, name_of_employee, name_of_department)
- **Projects** (Project_id, name, number_of_employee, start_date, end_date, budget, company_id, description).

Step 3: Identify the Relationship needed

1. Employees **work_for** Department
2. Employee **Manages** Department
3. Employees **work_on** Project
4. Department **Controls** Project
5. Manager **Leading** Department

Step 4: Identify the Cardinality Ratio and Participation

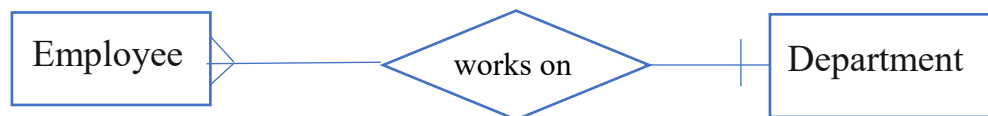
1. Company to Department: One-to-Many

Participation: Total for Department



2. Employee to Department: Many-to-One

Participation: Total for Employee, Partial for Department

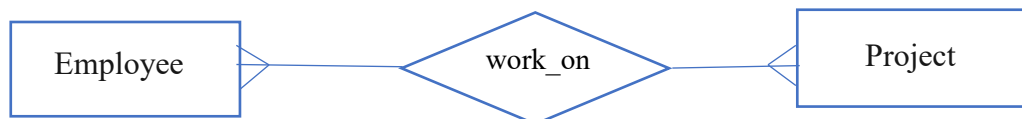


3. Department to project: One-to-Many



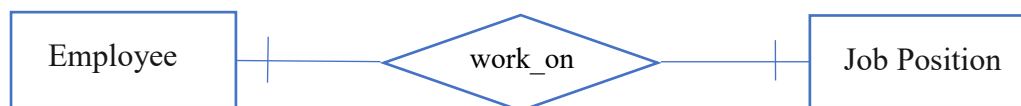
4. Employee to Project: Many-to-Many

Participation: Total for both Employee and Project

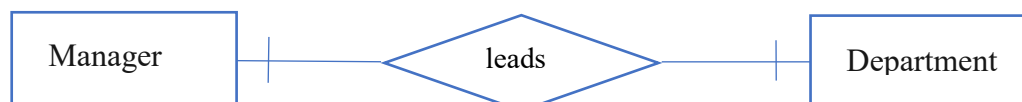


5. Employee to Job Position: One-to-One

Participation: Total for Job Position



6. Manager leads Department: One to One



Step 5: Draw the Diagram

